

# A self-organising Thucydides trap: growth, predictability, alliances

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*Subproject `selforg_alliances/`. Independent of the lattice Gillespie model in the parent repo; built from scratch on the brief’s minimal principles. Code: `thucydides_selforg.py`, figures `make_figures_selforg.py`, controls `experiments_selforg.py`.*

## 1 Aim

Reproduce the Thucydides trap—a rising challenger and an established power driven to systemic war—from **minimal, self-organising** principles, with two ingredients in the lead role:

1. **Predictability.** Each state forecasts the *growth* of the others and acts (war / ally / wait) on the forecast. Predictability is **endogenous and under selection**: states that fail to anticipate are destroyed.
2. **Balancing alliances.** States ally to avoid being attacked by a much stronger power. The hoped-for outcome is a **bimodal (two-bloc) structure** as the configuration from which the big war breaks out.

## 2 Model

$N = 100$  states, continuous strength  $S_i(t) \geq 0$ . Time advances in *rounds* of length  $\Delta = n_{\text{sub}} dt$ ; within each round the growth SDE is integrated by  $n_{\text{sub}}$  Euler steps, then forecasts are refreshed and the alliance, war, and predation steps are applied in turn. Figure ?? summarises the four mechanisms.

(i) **Resource-bounded stochastic growth.** Itô, Euler–Maruyama,

$$dS_i = \left[ \gamma_i S_i \left(1 - \frac{S_i}{c_i}\right) R(t) - \delta S_i \right] dt + \sigma \sqrt{S_i} dW_i, \quad R(t) = \max\left(0, 1 - \frac{\sum_j S_j}{K}\right). \quad (1)$$

Heterogeneous intrinsic rate  $\gamma_i$  and per-state capacity  $c_i$  (log-normal “territory” → a great-power/minor-power hierarchy). The global factor  $R$  is the shared-resource ceiling: no infinite growth. Demographic  $\sqrt{S}$  noise (CIR type) keeps  $S \geq 0$ .

(ii) **Predictability.** “Predictability” has two parts: a *trend estimate* that the whole system shares, and an *idiosyncratic forecast accuracy* that differs between states and is the trait under selection.

**Trend estimate.** At the end of each round, every state's realised slope  $r_j = [S_j(t) - S_j(t - \Delta)]/\Delta$  feeds an exponential moving average

$$\hat{S}_j \leftarrow (1 - \alpha) \hat{S}_j + \alpha r_j, \quad \alpha = \Delta/\tau_{\text{ema}}, \quad (2)$$

which smooths the demographic noise to recover the underlying growth rate. The projection a horizon  $h$  ahead is the linear extrapolation  $\hat{S}_j = S_j + h \hat{S}_j$  (Fig. ??A). (*Note the two distinct time scales:  $\Delta = n_{\text{sub}} dt = 1$  is the round length—the clock on which the SDE is integrated and decisions are taken—whereas  $\tau_{\text{ema}} = 8$  is the memory of the slope filter, i.e. how many past rounds are averaged; here  $\alpha = \Delta/\tau_{\text{ema}} = 1/8$ .)*

**Forecast accuracy.** Each state carries an accuracy  $a_i \in (0, 1]$ . When  $i$  needs another state's projected strength it reads it with multiplicative Gaussian error,

$$\tilde{S}_j^{(i)} = \max\left(0, \hat{S}_j(1 + \varepsilon_j^{(i)})\right), \quad \varepsilon_j^{(i)} \sim \mathcal{N}(0, \eta_i^2), \quad \eta_i = \sigma_p(1 - a_i). \quad (3)$$

$a_i \rightarrow 1$  is a near-perfect forecaster ( $\eta \rightarrow 0$ );  $a_i \rightarrow 0$  sees projected strengths blurred by noise of scale  $\sigma_p$  and can misjudge who is rising or who is stronger (Fig. ??B). *Every decision below—which threat to balance, which war to join—uses  $i$ 's own perceived quantities  $\tilde{S}^{(i)}$ , never the true values.*

**Self-organisation.**  $a_i$  is a heritable trait. When a state is eliminated it respawns as the offspring of a survivor drawn with probability  $\propto S$ , inheriting  $a$  (and  $\gamma, c$ ) with a small Gaussian mutation. Two forces make  $a_i$  adaptive: a good forecaster (a) joins winning coalitions and avoids losing ones, and (b) is hard to catch unprepared—predation hazard  $\propto (1 - 0.85 a_i)$  (mechanism iv). The blind are eaten, the foresighted survive and reproduce, so  $\bar{a}$  rises *endogenously*: predictability is selected for rather than assumed.

**(iii) Balancing alliances.** The states are partitioned into blocs (an alliance is a connected label set; initially all  $N$  states are singletons). Each round, every state independently reconsiders its membership with probability  $\rho$ . A reconsidering state  $i$  first forms the perceived strength of each bloc  $B$  from its own forecasts,

$$\tilde{\Sigma}_B^{(i)} = \sum_{j \in B} \tilde{S}_j^{(i)}, \quad (4)$$

and identifies its *top threat* as the strongest bloc it is not in,  $T = \arg \max_{B \neq b(i)} \tilde{\Sigma}_B^{(i)}$ . It then applies four rules, in order (Fig. ??C):

**R1 Salience.** If the threat does not clearly outweigh  $i$ ,  $\tilde{\Sigma}_T^{(i)} < \theta \tilde{S}_i^{(i)}$ , then no alliance is justified: if currently allied,  $i$  leaves to a singleton (prob. 0.6). This is the crux of the emergent topology—when power is diffuse *no* threat is salient and the system stays multipolar, whereas a single risen power is a salient threat to *many* states, who therefore converge on the *same* balancing coalition  $\Rightarrow$  bipolarity.

**R2 Dissolution.** Else, if  $i$ 's own bloc already dwarfs the threat,  $\tilde{\Sigma}_{b(i)}^{(i)} > 1.8 \tilde{\Sigma}_T^{(i)}$ , the alliance has outlived its purpose (and dilutes the spoils):  $i$  defects (prob. 0.6). R1+R2 are what let winning coalitions *fragment* after a war.

**R3 Balancing.** Otherwise  $i$  joins the bloc  $B^*$  (excluding the threat  $T$ ) that best matches the threat from above,

$$B^* = \arg \max_{B \neq T} \left[ -|\Sigma_B^{\text{side}} - \tilde{\Sigma}_T^{(i)}| + 0.25 \Sigma_B^{\text{side}} \right], \quad \Sigma_B^{\text{side}} = \tilde{\Sigma}_B^{(i)} + \tilde{S}_i^{(i)} [B \neq b(i)], \quad (5)$$

i.e. pick the camp whose strength (with  $i$  added) most nearly covers the threat, with a mild preference for the larger camp. This is textbook balance-of-power: aggregate just enough weight to deter the strongest player.

**R4 Bandwagon.** If instead the threat is overwhelming,  $\tilde{\Sigma}_T^{(i)} > 2.5 \tilde{S}_i^{(i)}$ , then with small probability  $p_{\text{bw}}$  the state *joins*  $T$  (a weak state buying protection from a hegemon rather than resisting it).

No geography or ideology is imposed; the entire bloc structure—its size, balance, and number—is emergent from these local rules acting on perceived strengths.

**Alliances in war.** R1–R4 set *standing* alignments; whether an ally actually fights is a separate decision taken at war time. When lead states  $\ell_A, \ell_B$  go to war, each ally  $k$  of a lead  $\ell$  joins with probability  $[1 + e^{-u_k}]^{-1}$ , where

$$u_k = g\left(P_k - \frac{1}{2}\right) + 0.4 \tanh(\tau_k - 1), \quad P_k = \frac{(\sigma_k^{\text{own}})^q}{(\sigma_k^{\text{own}})^q + (\sigma_k^{\text{opp}})^q}, \quad \tau_k = \frac{\sigma_k^{\text{opp}}}{\tilde{S}_k^{(k)}}, \quad (6)$$

with  $\sigma_k^{\text{own}} = \tilde{S}_\ell^{(k)} + \tilde{S}_k^{(k)}$  and  $\sigma_k^{\text{opp}} = \tilde{S}_{\ell_{\text{opp}}}^{(k)}$ : ally  $k$  joins when its *own* forecast says the war is winnable ( $P_k > \frac{1}{2}$ ) and the opposing side threatens it ( $\tau_k > 1$ ). A low- $a_k$  ally, mis-reading  $P_k$ , joins losing wars—another channel through which predictability pays.

**(iv) War—the Thucydides trigger.** Any sufficiently balanced pair of blocs (min / max strength  $> 0.25$ ) may fight, with hazard

$$\lambda = w \cdot \underbrace{\frac{\min(S_A, S_B)}{\max(S_A, S_B)}}_{\text{parity}} \cdot \underbrace{[1 + e^{-\kappa c}]^{-1}}_{\text{rising-challenger}}, \quad c = \frac{h(\dot{S}_{\text{chall}} - \dot{S}_{\text{rul}})}{S_A + S_B}, \quad (7)$$

i.e. war fires preferentially at **parity** *and* **when the weaker side’s projected growth is closing the gap** (preventive war against a riser;  $\hat{S}$  are the estimated slopes of the two lead states). Once initiated between the two lead states, allies join by the war-time rule given in (iii). Outcome is Lanchester,  $P(A) = S_A^q / (S_A^q + S_B^q)$ . The losing side pays `loss_lose`, the winner `loss_win`. **Predation:** a weak *unaligned* state next to a strong bloc is conquered with hazard  $\propto (1 - 0.85 a_i)$ —the blind get eaten, the foresighted ally in time and survive. Eliminated states respawn weak as singletons  $\rightarrow$  continual turnover.

**Parameters (baseline).** Growth:  $K=1500$ ,  $\delta=0.035$ ,  $\gamma \in [0.06, 0.16]$ ,  $\sigma=0.16$ ,  $c_i \sim \text{LogN}(\bar{c}=12, 0.6)$ , round  $\Delta = 1$  ( $n_{\text{sub}}=5$ ,  $dt=0.2$ ). Predictability:  $\tau_{\text{ema}}=8$ ,  $h=25$ ,  $\sigma_p=0.35$ . Alliances:  $\rho=0.15$ ,  $\theta=1.3$ ,  $p_{\text{bw}}=0.10$ . War/predation:  $w=0.05$ ,  $\kappa=12$ ,  $q=1.6$ ,  $g=6$ , `loss_lose`=0.45, `loss_win`=0.12, predation rate 0.05. Runs: 4000 rounds.

### 3 Results

Aggregates over **8 seeds** (mean  $\pm$  sd), with two mechanism controls:

| Quantity                                   | Baseline                                  | Reading                                    |
|--------------------------------------------|-------------------------------------------|--------------------------------------------|
| Effective # blocs $N_{\text{eff}}$         | <b><math>1.99 \pm 0.03</math></b>         | system self-organises to <b>bipolarity</b> |
| Bloc balance $1 -  f_1 - f_2 /(f_1 + f_2)$ | <b><math>0.92 \pm 0.06</math></b>         | the two camps are <b>strength-balanced</b> |
| Gini( $S$ )                                | $0.40 \pm 0.08$                           | persistent great/minor-power hierarchy     |
| Predictive accuracy $\bar{a}$              | <b><math>0.58 \rightarrow 0.67</math></b> | <b>selected upward</b> (init sd only 0.03) |
| Crossover fraction, systemic wars          | <b><math>0.44 \pm 0.07</math></b>         | systemic wars are rising-challenger wars   |
| Crossover fraction, small wars             | <b><math>0.11 \pm 0.03</math></b>         | — $4\times$ enrichment                     |
| War-size distribution                      | heavy-tailed ( $\approx 4$ decades)       | skirmishes $\rightarrow$ systemic wars     |

### Main outcomes.

1. **Bipolarity is the robust attractor.** Pure self-organised balancing drives  $N_{\text{eff}} \rightarrow 2$  with balanced camps, every seed. The bimodal alliance structure the brief hoped for *is* the generic state of the system—it is the standing backdrop, not a rare precursor.
2. **Systemic wars are Thucydidean.** Within the bipolar structure, the *big* wars are timed by a **projected power crossover** (rising challenger): 44% of systemic wars carry a crossover vs 11% of skirmishes, and systemic wars fire *below* full parity (0.75 vs 0.90) —i.e. **preventively, before the challenger arrives**. Control  $\kappa = 0$  (trigger off) roughly halves the systemic-war crossover fraction ( $0.44 \rightarrow 0.25$ ).
3. **Predictability self-organises.** Accuracy rises  $0.58 \rightarrow 0.67$  under selection. Removing the explicit foresight $\rightarrow$ survival coupling (accuracy-neutral predation) cuts the gain ( $\rightarrow 0.64$ ) but does not erase it: foresight also pays through better war-join decisions. Predictability is thus selected through *several* channels, dominated by avoiding predation.
4. **Heavy-tailed war sizes + rise-and-fall cycles** emerge without tuning (panels a, c, f): total power runs as a sawtooth—grow under the ceiling, systemic war resets it, regrow.

## 4 Interpretation & novelty (literature flags)

The pieces map onto known results—important to be honest about what is *not* new before claiming PRL novelty:

- **Bipolarity from balancing** is essentially Galam’s spin-glass coalition result (Galam, *Physica A* 1998/2002; Axelrod–Bennett “landscape theory” 1993, which retrodicted the WWII line-up). That balancing  $\rightarrow$  two blocs is *expected*.
- **Power-law war sizes from a self-organising geopolitical model** is Cederman, *APSR* 2003 (“Modeling the size of wars”, SOC)—and is exactly the phenomenology of the parent repo’s lattice model. Not new on its own.
- **Preventive war against a riser** is power-transition theory (Organski–Kugler 1980; Lemke), popularised as the “Thucydides trap” (Allison 2017). Standard in IR, comparatively fresh as a *dynamical-systems trigger*.

**What looks genuinely new** is the *combination*: a continuous-time **Langevin** geopolitics in which (a) **predictability is an endogenous trait under selection**, not a fixed rationality assumption, and (b) systemic wars are timed by a **projected-growth crossover** inside a self-organised bipolar

structure. The selection-for-foresight element—*states are pushed to become predictable because the unpredictable are eaten*—is the part I have not seen in the physics or IR-ABM literature.

## 5 Limitations of this v0

- Bipolarity is *too* robust ( $N_{\text{eff}}$  pinned at 2): with no latent geography/ideology, balancing trivially collapses to two blocs, so the multipolar $\leftrightarrow$ bipolar *transition* (arguably the most interesting object) is absent. The model cannot currently show bipolarity *emerging* before a war because it is bipolar essentially always.
- The “big-war” cut ( $\geq 10\%$  of world power destroyed) is a threshold, not an intrinsic scale; the heavy tail should be characterised as a distribution, not binarised.
- Selection on  $a_i$  is real but shallow ( $\Delta \approx 0.1$ ); the fitness gradient is parity-limited and partly hand-installed via the predation coupling.
- Alliances are connected-component labels; no notion of a state in two alliances, of alliance *reliability*, or of reputation.

## 6 How to proceed (ranked)

**A. Make multipolar $\leftrightarrow$ bipolar a real transition (highest value).** Add a latent position  $x_i$  (1-D ring or 2-D)—geography/ideology. Let balancing prefer *near* partners and threat fall off with distance. Then regional (multipolar) blocs are the low-tension phase, and a single rising hegemon overrides locality to force **global bipolarisation**. The order parameter  $N_{\text{eff}}$  then *moves*, and one can ask the sharp question: **does  $N_{\text{eff}} \rightarrow 2$  predict the next systemic war?** This converts finding (1) from “always bipolar” into a genuine precursor—the PRL hook.

**B. Find the sharp single claim + scaling law (PRL discipline).** Two candidates:

- *Predictability  $\leftrightarrow$  stability scaling.* Sweep the selection strength / mean accuracy  $\bar{a}$  and measure systemic-war rate or mean inter-war time  $\tau(\bar{a})$ . Hypothesis: better collective foresight **lengthens the long peace** but **enlarges** the eventual war (preventive wars launched earlier, at higher stakes)—a predictability–severity trade-off with a clean exponent. Novel and PRL-shaped.
- *Critical line in  $(w, \kappa)$*  separating a “frozen bipolar long peace” from a “churning” phase, with diverging fluctuations / power-law inter-war times at the boundary (SOC vs tuned criticality—test which).

**C. Analytics.** A 2-population mean-field (two blocs of strengths  $S_A, S_B$  under the growth SDE + crossover-triggered reset) is low-dimensional and likely tractable: a Langevin/Fokker–Planck for the gap  $g = S_A - S_B$  with an absorbing “war” boundary at projected crossover gives a **first-passage** prediction for the inter-war time distribution—the analytic backbone a PRL wants. Small- $N$  and weak-noise limits check the simulation.

**D. Confront data (reuse, don't rebuild).** The parent repo already loads COW / NMC / MID. Two cheap tests: (i) does the empirical major-power system sit near  $N_{\text{eff}} \approx 2$  before systemic wars (1914, 1939, Cold War)? (ii) is relative-CINC *slope* (rising challenger), not level, the better war predictor—the projected-crossover claim, directly testable on NMC time series.

**E. Cleanups.** Characterise the war-size tail (MLE power-law exponent +  $x_{\text{min}}$ , Clauset–Shalizi–Newman) instead of the 10% cut; add a reliability/defection parameter to alliances; check robustness of selection to mutation rate.

**Recommended first move.** **A + the  $\tau(\bar{a})$  scaling in B**—together they turn the two lead ingredients (alliances, predictability) into one measurable precursor and one scaling law, which is enough spine for a PRL.

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**Part II (the v2 model: dynamic growth, deliberation by Monte-Carlo rollout, war-driven absorption of the loser, and the multipolar $\leftrightarrow$ bipolar transition with Options A & B) has moved to its own self-contained document, NOTE.v2.tex.**

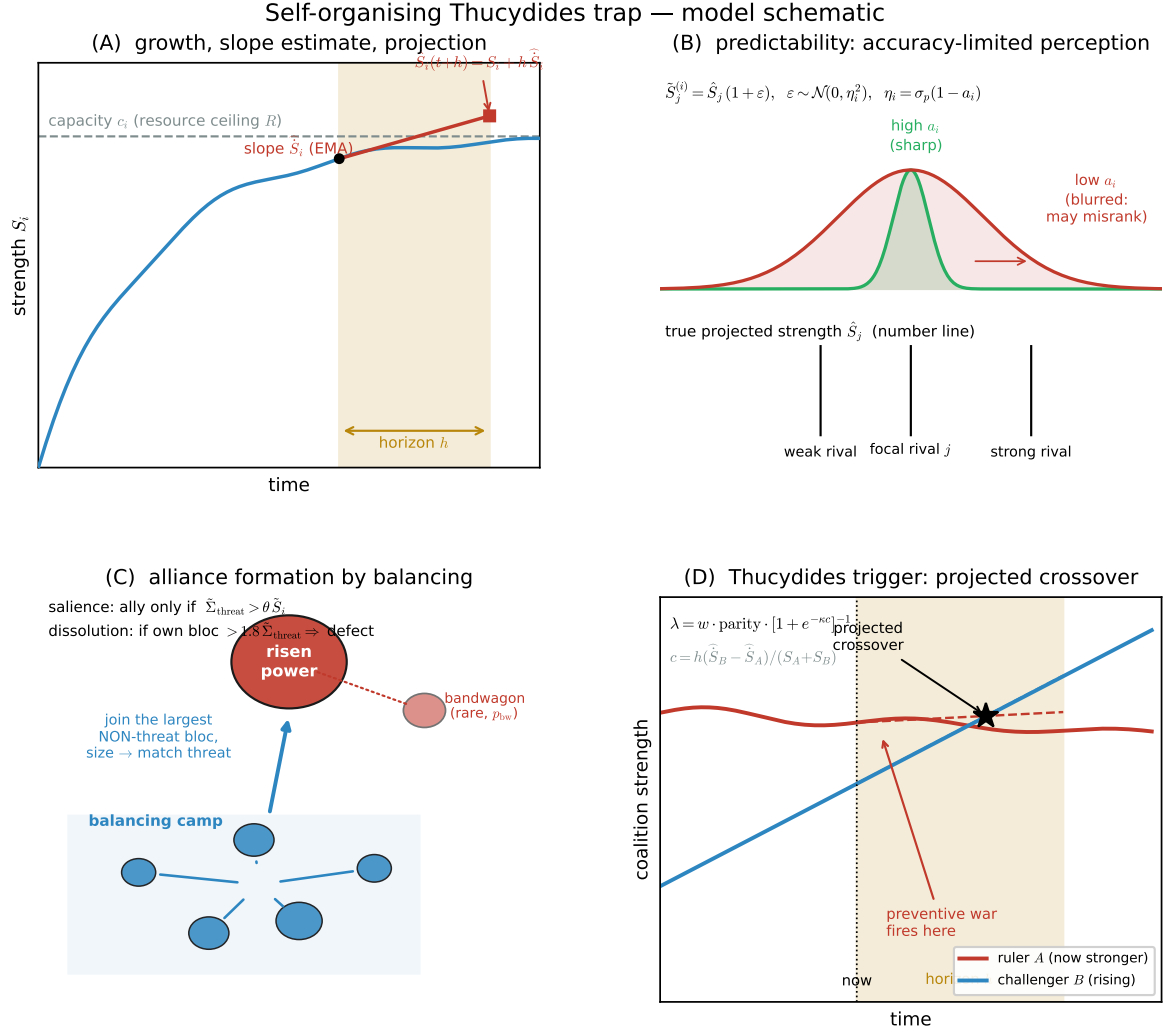


Figure 1: Model schematic. **(A)** Each state grows under a resource ceiling; its trend  $\hat{S}_i$  is estimated online and projected a horizon  $h$  ahead to  $\hat{S}_i(t+h)$ . **(B)** A state perceives others' projected strengths through accuracy-limited noise  $\eta_i = \sigma_p(1 - a_i)$ : high  $a_i$  gives a sharp estimate, low  $a_i$  a blurred one that can misrank rivals. **(C)** Balancing: a state allies only against a *salient* threat, joining the largest non-threat bloc to match it; it defects when no threat remains; it occasionally bandwagons onto an overwhelming power. **(D)** War ignites when a rising challenger's projected trajectory is set to cross the ruler's within the horizon—a preventive war fired *before* the crossing.

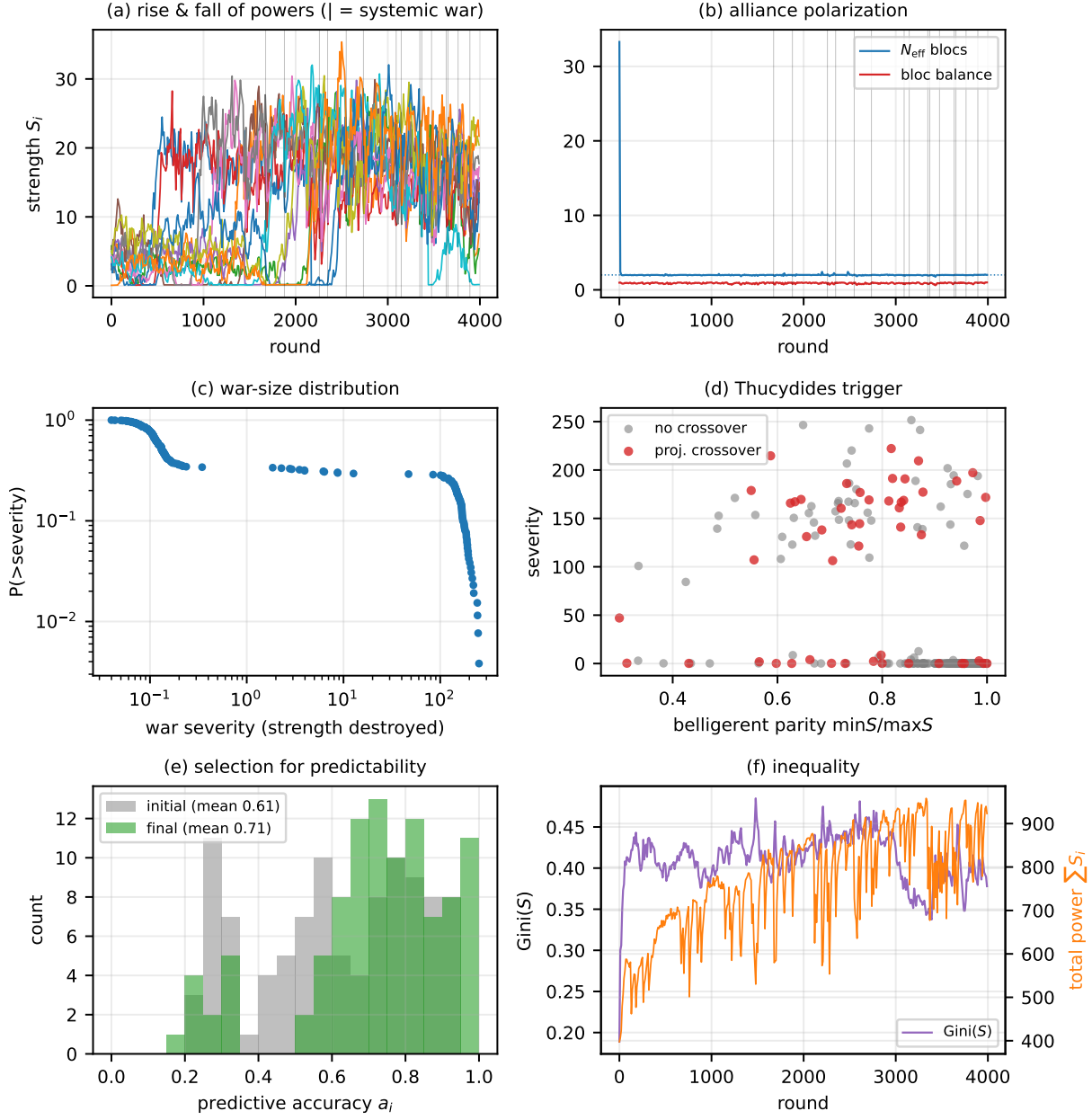


Figure 2: (a) rise-and-fall of the strongest powers, systemic wars marked; (b) alliance polarization  $N_{\text{eff}}$  and bloc balance; (c) war-size CCDF; (d) Thucydides trigger—severity vs belligerent parity, red = projected crossover; (e) selection on predictive accuracy (initial→final); (f) inequality and total power (sawtooth = grow/war/regrow).